

CHEATSHEET · QUICK REFERENCE

Lab 21 — Multi-Factor Authentication with TOTP (PAM)

Add a TOTP second factor to SSH via PAM, generate codes from the CLI, keep scratch codes for recovery, and apply the factor-category test for real MFA.

Provision the TOTP secret

```
sudo -u mfauser google-authenticator -t -d -f -r 3 -R 30 -w 17 -Q UTF8
Generate the user's TOTP secret + QR

head -3 /home/mfauser/.google_authenticator
View secret + config
```

Wire PAM + sshd

```
echo 'auth required pam_google_authenticator.so nullok' > /etc/pam.d/sshd.mfa
PAM TOTP module

KbdInteractiveAuthentication yes
Allow keyboard-interactive prompts

AuthenticationMethods publickey,keyboard-interactive
Require key + code

systemctl restart ssh
Apply
```

Test and recover

```
oathtool --totp -b "$SECRET"
Generate the current 6-digit code

ssh -o PreferredAuthentications=keyboard-interactive,password mfauser@127.0.0.1
Login prompts for password then code

tail -5 /home/mfauser/.google_authenticator
One-time scratch codes
```

Factor categories

Factor	In this lab	Other examples
Know	SSH password	PIN
Have	TOTP from app	YubiKey, smart card
Are	(not used)	Fingerprint, iris